

Sanjay Das

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EDUCATION

University of Texas at Dallas <i>Ph.D. in Computer Engineering (GPA: 3.89/4.0) • Advisor: Dr. Kanad Basu</i>	Richardson, TX, USA Aug. 2022 – Dec. 2025
North Dakota State University <i>Master's in Electrical and Computer Engineering (GPA: 4.0/4.0) • Advisor: Dr. Sumitha George</i>	Fargo, ND, USA Aug. 2021 – Aug. 2022
RCC Institute of Information Technology <i>Bachelor's in Electrical Engineering (CGPA: 8.6/10)</i>	Kolkata, West Bengal, India Jun. 2015 – May 2019

EXPERIENCE

Postdoctoral Research Associate <i>Learning Systems Group, CSMD, CCSD, Oak Ridge National Laboratory</i> - Led the design and deployment of laboratory-scale AI systems for operational safety, compliance, and decision support. - Exploring state-of-the-art AI capabilities through research and development in agentic scientific discovery, autonomous hypothesis generation and verification.	Jan. 2026 – Present Oak Ridge, TN, USA
Graduate Research Assistant <i>TIES Research Lab, University of Texas at Dallas</i> - Devised Fault mitigation in DNNs using uncertainty quantification and multi-exit networks (DAC'24). - Enhanced the Security of DNNs by creating high-quality functional test patterns for detecting attacks (HOST'24). - Investigated the vulnerabilities of Binary Neural Networks to Adversarial Bit Flipping (TMLR'24). - Developed a novel graph learning-based approach for fault criticality analysis in E/E systems (DAC'24, DAC'25).	Aug. 2022 – Dec. 2025 Richardson, TX, USA
ORISE Fellow (Internship) <i>Oak Ridge National Laboratory</i> - Leverage AI techniques, including GenAI and reasoning LLMs, to develop a semi-automated Human-in-the-Loop (HITL) system for high-risk property identification. - Build an agentic AI workflow, towards forecasting hazardous events at DOE sites, conducting risk analysis based on past incidents to enhance safety.	Jan. 2025 – July 2025 Oak Ridge, TN, USA
Graduate Research/Teaching Assistant <i>North Dakota State University</i> Designed a binary converter using FeFETs, binary, and ternary In-Memory Computing algorithms using QAHE memories.	Aug. 2021 – July 2022 Fargo, ND, USA
System Engineer <i>Tata Consultancy Services</i> Developed tailored IFS application customizations, optimizing using C#, Azure, PL/SQL, and Oracle Database expertise.	Sep. 2019 – July 2021 Bhubaneswar, India

TECHNICAL SKILLS

Programming & Scripting: Python, C/C++, Verilog, SQL, PLSQL, Bash/TCL Scripting, HTML/CSS (Basics).
ML & DL Libraries: TensorFlow, PyTorch, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, NetworkX, DGL, Hugging Face Transformers, LangChain, LlamaIndex, spaCy, NLTK, Accelerate.
Hardware Design & EDA Tools: Synopsys Tools (HSPICE, VCS, Design Vision, Tetramax), Cadence Tools (Virtuoso, Xcelium), Quartus Prime, FPGA, Chisel.
Cloud & DevOps / ML Deployment: Microsoft Azure, GitHub, GitLab, Oracle database, Microsoft Visual Studio, PyCharm, WordPress, FastAPI, JupyterLab.
Techniques & Tools: Retrieval Augmented Generation, Fine-Tuning LLMs, LoRA, QLoRA, LangChain, Vector DB - FAISS, ChromaDB, Prompt Engineering, LangChain Agents, CrewAI, grounded reasoning, tool-augmented LLMs, Retrieval-Augmented Generation (RAG), multimodal learning, knowledge graphs, and multi-agent workflows.

PUBLICATIONS

- Das, Sanjay, et al. "Enhancing Operational Safety via Agentic Dialogue Hazard Identification Analysis," in Proceedings of the 27th Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL), 2026.
- Das, Sanjay, et al. "HARNESS: Human-Agent Risk Navigation and Event Safety System for Proactive Hazard Forecasting in High-Risk DOE Environments," in The 1st Workshop on Retrieval Augmented Generation for Report Generation (RAG-ReportGen) @ ACL 2026.
- Das, Sanjay, et al. "ORCHID: Orchestrated Retrieval-Augmented Classification with Human-in-the-Loop Intelligent Decision-Making for High-Risk Property," in The 1st Workshop on Retrieval Augmented Generation for Report Generation (RAG-ReportGen) @ ACL 2026.

- Das, Sanjay, et al. "AttentionBreaker: Adaptive Evolutionary Optimization for Unmasking Vulnerabilities in LLMs through Bit-Flip Attacks," in Transactions on Machine Learning Research (TMLR), 2025.
- Das, Sanjay, et al. "One Bit-Flip Away from Collapse: Security and Safety in AI Models and Accelerators," in 2025 First International Conference on Intelligent Computing and Systems at the Edge (ICEEdge), 2025.
- Das, Sanjay, et al. "CONCEAL: Covert NoC Exploitation in In-Memory Computing-based DNN Accelerators," in 2025 IEEE International Conference on Omni-layer Intelligent Systems (COINS), 2025.
- Das, Sanjay, et al., "Bit-by-Bit: Investigating the Vulnerabilities of Binary Neural Networks to Adversarial Bit Flipping." In Transactions on Machine Learning Research (TMLR) (2024).
- Kundu, Shamik, Sanjay Das, et al., "MENDNet: Just-in-time Fault Detection and Mitigation in AI Systems with Uncertainty Quantification and Multi-Exit Networks. " In Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC '24), 2024.
- Das, Sanjay, et al., "Analyzing and Mitigating Circuit Aging Effects in Deep Learning Accelerators. " 2024 IEEE 42nd VLSI Test Symposium (VTS).
- Choudhury, Navnil, Chaithanya Naik Mude, Sanjay Das, et al. "Crosstalk-induced Side Channel Threats in Multi-Tenant NISQ Computers." arXiv preprint arXiv:2412.10507 (2024).
- Das, Sanjay, et al., "Explainability to the Rescue: A Pattern-Based Approach for Detecting Adversarial Attacks. " in 2024 IEEE International Symposium on Hardware Oriented Security and Trust, 2024.
- Das, Sanjay, et al., "Graph Learning-based Fault Criticality Analysis for Enhancing Functional Safety of E/E Systems. " In Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC '24), 2024.
- Srivastava, Amisha, Sanjay Das, et al., "SCAR: Power Side-Channel Analysis at RTL Level." IEEE Transactions on Very Large-Scale Integration (VLSI) Systems (2024).
- Arunachalam, Ayush, Sanjay Das, et al., "Enhanced ML-Based Approach for Functional Safety Improvement in Automotive AMS Circuits," 2023 IEEE International Test Conference (ITC), Anaheim, CA, USA, 2023.
- Das, Sanjay, et al., "Bottlenecks in Secure Adoption of Deep Neural Networks in Safety-Critical Applications," 2023 IEEE 66th International Midwest Symposium on Circuits and Systems (MWSCAS), Tempe, AZ, USA, 2023.
- Das, Sanjay, et al., "Adaptable Multi-level Voltage to Binary Converter Using Ferroelectric FETs." In 2022 IEEE Computer Society Annual Symposium on VLSI (ISVLSI), pp. 116-121. IEEE, 2022.
- Das, Sanjay, et al., "Hardware realization of an intelligent medical image watermarking." International Journal of Nanoparticles (2020).

ACADEMIC SERVICES

Reviewer, IEEE Design & Test (D&T), 2026.

Reviewer, IEEE Transactions on Very Large Scale Integration Systems (TVLSI), 2026.

Reviewer, European Chapter of the Association for Computational Linguistics (EACL), 2026.

Reviewer, Asia and South Pacific Design Automation Conference (ASP-DAC) 2023.

Reviewer, IEEE International System-on-Chip Conference (SOCC), 2024.

Reviewer, International Conference on Computer-Aided Design (ICCAD), 2024.

Reviewer, VLSI Test Symposium (VTS), 2025.

STUDENTS MENTORED

- Irfan Yakub Ali Khan, 2024-2025

RELEVANT COURSES

Machine Learning • Natural Language Processing • Probability • Linear Algebra • Data Structures & Algorithms • Operating Systems • Computer Architecture • Data Science with R • Cybersecurity • Databases • Digital Logic Design • Differential Equations • Cryptocurrency

AWARDS AND ACHIEVEMENTS

- Placed **First** at the 14th ORPA Research Symposium at Oak Ridge National Laboratory for poster: "Intelligent Parameter Space Exploration for Enabling Agentic Cosmological Simulations."
- Received "**Mary and Richard Templeton Fellowship**" for the 2024-25 calendar year at the University of Texas at Dallas.
- Received "**Jonsson School IAC Graduate Fellowship**" for the 2023-24 calendar year at the University of Texas at Dallas.
- Placed third in the prestigious Hack@DAC competition at the Design Automation Conference (DAC), 2024.
- Finalist in Logic Locking and Embedded Security Challenge at Cyber Security Awareness Week, 2022.
- "Adaptable Multi-level Voltage to Binary Converter using Ferro-Electric FETs" received the "**Best Paper Award**" at ISVLSI 2022.